

Gaussian Bridges and Discrete Frontiers in Zero-Bias Occupancy

A resolution of the geometric occupancy conjecture,
with sharp summability and centering criteria

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Abstract

An explicit conjecture in the Fabius–Rvachev research material of **ProveIt** asks for a limit shape and Gaussian fluctuations of the allocation of iterated zero-bias transformations among geometric uniform summands. We prove that conjecture and strengthen it to convergence of the entire fluctuation profile in ℓ^1 . The limiting covariance is exactly one half of the multinomial covariance. The proof starts from an exact representation by independent Poisson variables, each conditioned to be odd, followed by conditioning the total allocation. A uniform lattice local limit theorem makes this last conditioning rigorous.

The same argument gives a universal ℓ^2 Gaussian bridge for every positive summable weight profile, and identifies $\sum_j \sqrt{c_j} < \infty$ as the necessary and sufficient condition for tightness in ℓ^1 . For power weights, centering at mc_j has a sharp transition at exponent two. For geometric weights, the last occupied scales have a different limit: an explicit independent discrete field depending on the fractional part of a logarithm. We obtain total-variation limits for the last occupied index and the number of occupied indices, all moments of the latter, exact phase averages, and asymptotic independence from the Gaussian bridge. Finally, the normalized profile satisfies a full ℓ^1 large deviation principle with rate twice relative entropy. The article includes complete proofs, reproducible exact and numerical checks, and a proposed research program. The results are mathematical proofs in a research manuscript; no Lean formalization or literature-wide priority is asserted.

Status and provenance. The resolved source statement is `conj:occupancy-shape` in *Zero-Bias Towers and Spectral Peeling in the Fabius–Rvachev System* [1]. The repository was inspected at commit `a21208b3ff14a07a4c8318dbf916d543acbef043`. The proofs below do not assume any of that report’s conjectures or its endpoint asymptotic claims.

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1 The problem and the results

1.1 Why these occupancies occur

The centered Fabius–Rvachev random series is

$$Y_{1/2} = \sum_{j \geq 0} 2^{-j-1} U_j, \quad U_j \text{ independent and uniform on } [-1, 1].$$

Its natural geometric extension is

$$Y_q = \sum_{j \geq 0} c_j U_j, \quad c_j = (1-q)q^j, \quad 0 < q < 1. \quad (1.1)$$

The random-series viewpoint belongs to the classical study of the compactly supported smooth function; see Arias de Reyna [6]. Zero bias is the distributional transformation characterized by

$$\mathbb{E}[Wg(W)] = \text{Var}(W)\mathbb{E}[g'(W^z)] \quad (1.2)$$

for a centered variable of positive finite variance. Goldstein and Reinert introduced and developed this transformation [2]. Symmetry is preserved, so it can be iterated on Y_q .

At level m , an exact representation of the transformed sum assigns an integer $K_j^{(m)}$ to each summand, with $\sum_j K_j^{(m)} = m$. Conditional on these allocations, the transformed digits are independent and the j th digit has law $2\text{Beta}(K_j^{(m)} + 1, K_j^{(m)} + 1) - 1$. The allocation law recorded in the repository is

$$\mathbb{P}(K^{(m)} = k) = \frac{1}{Z_m(c)} \prod_{j \geq 0} \frac{c_j^{2k_j}}{k_j!(2k_j + 1)!!}, \quad k_j \in \mathbb{N}_0, \quad \sum_j k_j = m. \quad (1.3)$$

Only finitely many k_j are nonzero. We rederive the normalization and its connection to zero bias in [section 2](#).

For one transformation, the location probabilities are proportional to c_j^2 . The conjecture predicts that after many transformations the *fraction* of allocations at location j approaches c_j itself, with Gaussian fluctuations of variance $mc_j(1 - c_j)/2$. The change from variance weights to amplitude weights is one of the structural features explained by the proof.

1.2 A complete answer to the stated conjecture

For a sequence x , write $\|x\|_1 = \sum_j |x_j|$.

Theorem 1.1 (Geometric occupancy conjecture and global strengthening). *Fix $0 < q < 1$, and use [equations \(1.1\)](#) and [\(1.3\)](#). Let $(Z_j)_{j \geq 0}$ be independent standard normal variables and set*

$$Z_* = \sum_{j \geq 0} \sqrt{c_j} Z_j, \quad G_j = \frac{\sqrt{c_j} Z_j - c_j Z_*}{\sqrt{2}}. \quad (1.4)$$

Then G is an ℓ^1 -valued random variable and

$$\frac{K^{(m)} - mc}{\sqrt{m}} \implies G \quad \text{in } \ell^1. \quad (1.5)$$

In particular,

$$\text{Cov}(G_i, G_j) = \frac{1}{2}(c_i \mathbf{1}_{\{i=j\}} - c_i c_j), \quad \sum_j G_j = 0 \quad \text{almost surely.} \quad (1.6)$$

The profiles $K^{(m)}/m$ converge to c in ℓ^1 , both in probability and in mean. More precisely,

$$\lim_{m \rightarrow \infty} \sqrt{m} \mathbb{E} \left\| \frac{K^{(m)}}{m} - c \right\|_1 = \frac{1}{\sqrt{\pi}} \sum_{j \geq 0} \sqrt{c_j(1 - c_j)}. \quad (1.7)$$

The repository conjecture asks only for each fixed finite vector. Thus [equation \(1.5\)](#) proves both of its assertions and adds uniform control over all scales. The expected total-variation distance between the two probability profiles is one half of the expectation in [equation \(1.7\)](#).

1.3 Further conclusions proved in this article

The global theorem is part of a broader structure.

- **A sharp functional threshold.** For every infinite positive profile c of total mass one, exact saddle centering gives an ℓ^2 Gaussian bridge. The square-root summability condition $\sum_j \sqrt{c_j} < \infty$ is exactly the criterion for ℓ^1 tightness ([theorems 4.3](#) and [4.4](#)).
- **A centering transition.** For $c_j = C(j+1)^{-\alpha}$, centering at mc_j is correct for $\alpha > 2$, leaves a nonzero Gaussian mean for $\alpha = 2$, and misses the scale of the displacement for $1 < \alpha < 2$ ([theorem 7.2](#)).
- **An explicit geometric frontier.** Around index $\log_{1/q}(2m(1-q))$, the allocation counts approach independent discrete laws. The maximum occupied index and the occupied count have phase-dependent lattice limits ([theorems 6.1](#) and [6.3](#)).
- **Exact occupied-count constants.** Its variance remains bounded, with phase average $(2 \log 2 - 1)/\log(1/q)$; its mean has an explicit periodic correction ([proposition 6.5](#)).
- **A global entropy law.** The entire probability profile has a large deviation principle in ℓ^1 with good rate $2D(p||c)$ ([theorem 8.1](#)).

1.4 Relation to existing methods and novelty

Conditioning independent components to represent a combinatorial law is an established method; Arratia and Tavaré give a systematic treatment [\[3\]](#). Infinite occupancy problems and their connections to regular variation have an extensive literature [\[4\]](#). The law [equation \(1.3\)](#) has its own factorial weights and is not an ordinary independent balls-in-boxes allocation. Its exact parity-conditioned representation determines the factor $1/2$ in the Gaussian covariance and the factor 2 in the entropy rate.

The contribution claimed here is the resolution of an explicitly identified repository conjecture, together with the extensions proved below. The classical tools and the source allocation identity are credited. A search of the relevant source, its README, and repository matches for the conjecture label found no proof of that conjecture at the inspected snapshot. This is a repository-relative assessment, not a proof of worldwide priority. All theorem statements in this article have ordinary mathematical proofs; numerical checks and informal independent reviews are supplementary evidence, not formal proof certificates.

2 The allocation law and its exact probabilistic representation

Unless a geometric profile is explicitly required, assume throughout that

$$c_j > 0 \quad (j \geq 0), \quad \sum_{j \geq 0} c_j = 1. \quad (2.1)$$

This infinite positive setting makes every fixed finite block have a complement of positive mass. Zero and finite-support cases are discussed in [remark 4.5](#).

2.1 Normalization by moments

Let $Y_c = \sum_j c_j U_j$ and $\mu_{2m}(c) = \mathbb{E}Y_c^{2m}$, with $\mu_0(c) = 1$. Absolute convergence gives $|Y_c| \leq 1$ almost surely, and its moment generating function is

$$M_c(t) = \prod_{j \geq 0} \frac{\sinh(tc_j)}{tc_j}. \quad (2.2)$$

The product converges locally uniformly for complex t : for small z , $\sinh z/z = 1 + O(z^2)$, and $\sum c_j^2 < \infty$.

Proposition 2.1 (Coefficient normalization). *The positive normalizing constant in [equation \(1.3\)](#) is finite and equals*

$$Z_m(c) = \frac{\mu_{2m}(c)}{(2m-1)!! m!} = [z^m] \prod_{j \geq 0} \frac{\sinh(c_j \sqrt{2z})}{c_j \sqrt{2z}}. \quad (2.3)$$

Here $(-1)!! = 1$. The expression involving $\sqrt{2z}$ is an entire power series in z , with no choice of square-root branch involved.

Proof. The identity

$$\frac{\sinh(c\sqrt{2z})}{c\sqrt{2z}} = \sum_{k \geq 0} \frac{2^k c^{2k} z^k}{(2k+1)!} = \sum_{k \geq 0} \frac{c^{2k} z^k}{k!(2k+1)!!}$$

gives the sum of the weights on extracting the coefficient. On the other hand, $M_c(\sqrt{2z}) = \sum_m 2^m \mu_{2m}(c) z^m / (2m)!$ and $(2m)! = 2^m m! (2m-1)!!$. Positivity of a single weight proves $Z_m(c) > 0$. Finite truncations and monotone convergence of the nonnegative coefficients justify the infinite coefficient identity. $\square$

2.2 Connection to the zero-bias tower

For an even moment generating function put $\mathcal{D} = t^{-1}d/dt$, interpreting the origin by continuity. Applying [equation \(1.2\)](#) to exponentials gives

$$M_{W^z}(t) = \frac{\mathcal{D}M_W(t)}{\mathcal{D}M_W(0)}.$$

By induction, the m th transformed law has moment generating function

$$\frac{\mathcal{D}^m M_W(t)}{\mathcal{D}^m M_W(0)}, \quad \mathcal{D}^m M_W(0) = \frac{\mathbb{E}W^{2m}}{(2m-1)!!}. \quad (2.4)$$

The operator $\mathcal{D}$ is a derivation, so its multinomial product rule applied to [equation \(2.2\)](#) yields the weights in [equation \(1.3\)](#). At an individual uniform digit, the k th transform has density proportional to $(1-u^2)^k$ on $[-1, 1]$: one integration by parts verifies the next zero-bias identity, and induction starts with the uniform density. This is the law $2\text{Beta}(k+1, k+1) - 1$. All distributions have compact support, so equality of their entire moment generating functions determines their laws. Finite truncation then gives the countably infinite statement.

This argument identifies an allocation in an exact mixture for each level. It does not require a particular chronological coupling of all levels m on one probability space.

2.3 The parity-conditioned variables

Definition 2.2. For $a > 0$, let B_a be the nonnegative integer-valued variable with

$$\mathbb{P}(B_a = k) = \frac{a^{2k+1}}{(2k+1)! \sinh a}, \quad k \in \mathbb{N}_0. \quad (2.5)$$

Equivalently, $2B_a + 1$ is a Poisson variable of parameter a , conditioned to be odd. For $t > 0$, construct independent $B_j(t) \sim B_{tc_j}$ and put $S(t) = \sum_j B_j(t)$.

Theorem 2.3 (Exact conditioning identity). *The sum $S(t)$ is finite almost surely, $\mathbb{P}(S(t) = m) > 0$ for every $m \geq 0$, and*

$$\mathcal{L}(K^{(m)}) = \mathcal{L}((B_j(t))_{j \geq 0} \mid S(t) = m) \quad \text{for every } t > 0. \quad (2.6)$$

Proof. Direct summation or logarithmic differentiation gives

$$\mu(a) := \mathbb{E}B_a = \frac{a \coth a - 1}{2} \leq \frac{a}{2}. \quad (2.7)$$

Thus $\mathbb{E}S(t) \leq t/2 < \infty$. Also

$$\prod_j \mathbb{P}(B_j(t) = 0) = \prod_j \frac{tc_j}{\sinh(tc_j)} > 0,$$

because the tail logarithms are $O(t^2 c_j^2)$. Setting $B_0(t) = m$ and all other coordinates to zero gives a positive-probability event with total m . For any finite-support k with total m , its product probability is

$$\left(\prod_j \frac{tc_j}{\sinh(tc_j)} \right) t^{2m} \prod_j \frac{c_j^{2k_j}}{(2k_j + 1)!}.$$

The first two factors are common to the conditioning event, and replacing $(2k_j + 1)!$ by $2^{k_j} k_j! (2k_j + 1)!!$ adds the common factor 2^{-m} . This proves the identity. $\square$

Remark 2.4 (The countable construction matters). The $B_j(t)$ are defined by their separate probability laws and then made independent. They are *not* obtained by conditioning one infinite ordinary-Poisson vector on the event that every coordinate is odd. The latter event has probability zero. The final event $S(t) = m$, in contrast, has strictly positive probability by the preceding proof.

3 Saddle means and a uniform local limit theorem

3.1 Elementary estimates and the unique saddle

Define

$$f(a) = 1 - \frac{2a}{e^{2a} - 1}, \quad D(t) = \sum_{j \geq 0} f(tc_j), \quad M(t) = \mathbb{E}S(t), \quad V(t) = \text{Var } S(t). \quad (3.1)$$

We use $f(0) = 0$ by continuity. The expansion at zero and the large- a behavior are

$$f(a) = a - \frac{a^2}{3} + O(a^4), \quad \mu(a) = \frac{a^2}{6} + O(a^4), \quad (3.2)$$

$$f(a) = 1 + O(ae^{-2a}), \quad \mu(a) = \frac{a-1}{2} + O(ae^{-2a}). \quad (3.3)$$

Lemma 3.1 (Bounds and saddle identity). *For $a > 0$,*

$$0 < f(a) \leq \min(a, 1), \quad v(a) := \text{Var } B_a = \frac{a \coth a - a^2 \text{csch}^2 a}{4} \leq \frac{a}{4}. \quad (3.4)$$

There is a unique $t_m > 0$ satisfying $M(t_m) = m$ for every $m \geq 1$, and

$$M(t) = \frac{t - D(t)}{2}, \quad D(t) = o(t), \quad V(t) \sim \frac{t}{4}, \quad t_m = 2m + D(t_m) \sim 2m. \quad (3.5)$$

If $\mu_j(t) = \mu(tc_j)$, then

$$\mu_j(t_m) - mc_j = \frac{c_j D(t_m) - f(t_m c_j)}{2}, \quad \|\mu(t_m) - mc\|_1 \leq D(t_m). \quad (3.6)$$

Proof. The inequalities $f(a) > 0$ and $f(a) < 1$ follow from $e^{2a} > 1 + 2a$. The inequality $f(a) \leq a$ is equivalent to $a \coth a \geq 1$, which follows from $\tanh a \leq a$. Differentiating the probability generating function, or using the exponential-family identity, gives $v(a) = a\mu'(a)/2$ and the formula in [equation \(3.4\)](#). The upper bound is equivalent to

$$\sinh a \cosh a - \sinh^2 a \leq a,$$

whose left side is $(1 - e^{-2a})/2 \leq a$. Strict positivity of $v(a)$ follows because B_a is nondegenerate.

The series of means and variances converge locally uniformly by these bounds and their small- a expansions. Consequently M is continuous and strictly increasing, with $M'(t) = 2V(t)/t > 0$ and $M(t) \rightarrow 0$ as $t \downarrow 0$. For each fixed j , $f(tc_j)/t \rightarrow 0$, while $f(tc_j)/t \leq c_j$. Dominated convergence proves $D(t) = o(t)$ and hence $M(t) \rightarrow \infty$. Similarly $v(tc_j)/t \rightarrow c_j/4$ and is bounded by $c_j/4$, proving $V(t)/t \rightarrow 1/4$. This proves existence, uniqueness, and [equation \(3.5\)](#). Finally $\mu(tc_j) = (tc_j - f(tc_j))/2$, so substituting $t_m - 2m = D(t_m)$ gives the exact centering difference; summing the triangle inequality gives its norm bound. $\square$

The quantities $\mu_j(t_m)$ are the means of the *independent* variables before conditioning. They need not equal $\mathbb{E}K_j^{(m)}$. The term “saddle centering” will always refer to these explicitly defined means, not to an asserted formula for the conditional means.

3.2 A global Fourier bound

The characteristic function of B_a is

$$g_a(u) = \mathbb{E}e^{iuB_a} = e^{-iu/2} \frac{\sinh(ae^{iu/2})}{\sinh a}. \quad (3.7)$$

Although its numerator has this parity origin, B_a itself has lattice span one: every nonnegative integer has positive mass.

Lemma 3.2 (One coordinate controls the Fourier integral). *For $a \geq 1$ and $|u| \leq \pi$,*

$$|g_a(u)| \leq C_0 \exp\left(-\frac{au^2}{2\pi^2}\right), \quad C_0 = \frac{2}{1 - e^{-2}}. \quad (3.8)$$

Consequently there is an absolute constant C_1 such that

$$\sup_k \mathbb{P}(B_a = k) \leq \min(1, C_1 a^{-1/2}) \quad (a > 0). \quad (3.9)$$

Proof. For real x, y , $|\sinh(x + iy)|^2 = \sinh^2 x + \sin^2 y \leq \cosh^2 x$. On $|u| \leq \pi$, $\cos(u/2) \geq 0$, so

$$|\sinh(ae^{iu/2})| \leq e^{a \cos(u/2)}, \quad \sinh a \geq \frac{1}{2}(1 - e^{-2})e^a.$$

Also $1 - \cos(u/2) \geq u^2/(2\pi^2)$. These give [equation \(3.8\)](#). Fourier inversion and the Gaussian integral give [equation \(3.9\)](#) for $a \geq 1$; increasing C_1 covers $0 < a < 1$ by the trivial bound one. $\square$

3.3 The local limit theorem

For a fixed nonempty set $A \subseteq \mathbb{N}_0$, finite or infinite, set

$$S_A(t) = \sum_{j \in A} B_j(t), \quad \mu_A(t) = \sum_{j \in A} \mu_j(t), \quad C_A = \sum_{j \in A} c_j.$$

Theorem 3.3 (Uniform lattice local limit theorem). *For every such A ,*

$$\sup_{k \in \mathbb{Z}} \left| \sqrt{t} \mathbb{P}(S_A(t) = k) - \sqrt{\frac{2}{\pi C_A}} \exp\left(-\frac{2(k - \mu_A(t))^2}{t C_A}\right) \right| \rightarrow 0. \quad (3.10)$$

In particular,

$$\mathbb{P}(S(t_m) = m) \sim \sqrt{\frac{2}{\pi t_m}} \sim \frac{1}{\sqrt{\pi m}}. \quad (3.11)$$

Proof. First fix j . In equation (3.7), take $u = w/\sqrt{t}$ and $a = tc_j$. The identity $\log \sinh z = z - \log 2 + \log(1 - e^{-2z})$ applies for positive real part. The last logarithm is exponentially small uniformly on this scaled argument. After centering, the logarithm of the characteristic function is

$$tc_j \left(e^{iw/(2\sqrt{t})} - 1 - \frac{iw}{2\sqrt{t}} \right) + o(1) \rightarrow -\frac{c_j w^2}{8}.$$

Thus $(B_j(t) - \mu_j(t))/\sqrt{t} \Rightarrow N(0, c_j/4)$. Independence gives the corresponding result for each finite sum. The variance of a remaining tail divided by t is at most one quarter of its c -mass, by equation (3.4); approximating A by finite sets therefore gives

$$\frac{S_A(t) - \mu_A(t)}{\sqrt{t}} \Rightarrow N(0, C_A/4). \quad (3.12)$$

Choose a fixed $j_0 \in A$. All other characteristic factors have modulus at most one. By lemma 3.2, after $u = w/\sqrt{t}$ the centered Fourier kernel of $S_A(t)$ is bounded in modulus by $C_0 \exp(-c_{j_0} w^2/(2\pi^2))$ on $|w| \leq \pi\sqrt{t}$. Extend that kernel by zero outside this interval. Equation (3.12) gives pointwise convergence to $e^{-C_A w^2/8}$, and dominated convergence gives convergence in $L^1(\mathbb{R})$. Fourier inversion now yields equation (3.10). Uniformity in k follows because the absolute difference of the inversions is bounded by the L^1 difference of their kernels, independently of the oscillating factor. The full set $A = \mathbb{N}_0$ has $C_A = 1$ and mean m at t_m , proving equation (3.11). $\square$

An ordinary central limit theorem alone would not justify conditioning on $S(t_m) = m$, whose probability tends to zero. The local theorem supplies both its precise mass and the likelihood ratios needed next.

4 Gaussian bridges in sequence spaces

4.1 Conditioning a fixed block

Proposition 4.1 (Finite-coordinate bridge). *For every fixed finite $I \subset \mathbb{N}_0$,*

$$\left(\frac{K_j^{(m)} - \mu_j(t_m)}{\sqrt{m}} \right)_{j \in I} \Rightarrow N\left(0, \frac{1}{2}(\text{diag}(c_I) - c_I c_I^T)\right). \quad (4.1)$$

Proof. Let $R = \mathbb{N}_0 \setminus I$ and $C_R = \sum_{j \in R} c_j > 0$. At $t = t_m$, the independent block's likelihood ratio under conditioning is

$$r_t(k) = \frac{\mathbb{P}(S_R(t) = m - \sum_{j \in I} k_j)}{\mathbb{P}(S(t) = m)}.$$

Set $z_j = (k_j - \mu_j(t))/\sqrt{t}$. Since the total mean is m , the centered argument in the numerator is $-\sqrt{t} \sum_{j \in I} z_j$. The uniform local theorem gives, uniformly in k ,

$$r_t(k) = C_R^{-1/2} \exp\left(-\frac{2(\sum_{j \in I} z_j)^2}{C_R}\right) + o(1). \quad (4.2)$$

These ratios are uniformly bounded. The independent block converges to a normal vector with covariance $\text{diag}(c_I)/4$. Hence its conditional limit has the normal density multiplied by the bounded continuous factor in [equation \(4.2\)](#). Combining the quadratic forms, or applying the rank-one inverse formula, gives covariance $(\text{diag}(c_I) - c_I c_I^T)/4$ on the $\sqrt{t}$ scale. Finally $t_m/m \rightarrow 2$, proving [equation \(4.1\)](#). $\square$

4.2 A bound uniform in the omitted coordinate

Lemma 4.2 (Uniform marginal domination). *There is a constant $C_c < \infty$ such that, for all sufficiently large m , every $j \geq 0$, and every nonnegative function u on $\mathbb{N}_0$,*

$$\mathbb{E}u(K_j^{(m)}) \leq C_c \mathbb{E}u(B_j(t_m)). \quad (4.3)$$

In particular,

$$\mathbb{E}(K_j^{(m)} - \mu_j(t_m))^2 \leq C_c t_m c_j / 4. \quad (4.4)$$

Proof. If $j \neq 0$, the complementary sum retains coordinate 0; if $j = 0$, it retains coordinate 1. A convolution's maximal atom is no larger than that of any factor. By [equation \(3.9\)](#), therefore,

$$\sup_{j,k} \mathbb{P}(S(t) - B_j(t) = k) \leq C t^{-1/2},$$

with a constant depending only on c_0, c_1 . Divide this by [equation \(3.11\)](#) in the exact conditional marginal formula. Then apply [equation \(3.4\)](#) to obtain [equation \(4.4\)](#). $\square$

4.3 A universal Hilbert-space limit

The series Z_* in [equation \(1.4\)](#) converges in L^2 and almost surely, since its independent summands have total variance one. The resulting G satisfies

$$\mathbb{E}\|G\|_2^2 = \frac{1}{2} \left(1 - \sum_j c_j^2\right) < \infty. \quad (4.5)$$

Theorem 4.3 (Universal ℓ^2 bridge). *Under [equation \(2.1\)](#), without additional tail assumptions,*

$$\frac{K^{(m)} - \mu(t_m)}{\sqrt{m}} \Rightarrow G \quad \text{in } \ell^2. \quad (4.6)$$

Proof. The vectors have the finite-dimensional limits in [proposition 4.1](#). By [equation \(4.4\)](#),

$$\mathbb{E} \sum_{j>J} \frac{|K_j^{(m)} - \mu_j(t_m)|^2}{m} \leq C \sum_{j>J} c_j \rightarrow 0$$

uniformly for large m . The same tail bound holds for G by its covariance. Approximating both vectors by their first $J+1$ coordinates proves convergence against bounded Lipschitz functions on ℓ^2 , and hence weak convergence. $\square$

4.4 The sharp ℓ^1 criterion

Theorem 4.4 (Necessary and sufficient square-root summability). *The family $(K^{(m)} - \mu(t_m))/\sqrt{m}$ is tight in ℓ^1 if and only if*

$$\sum_{j \geq 0} \sqrt{c_j} < \infty. \quad (4.7)$$

Under this condition,

$$\frac{K^{(m)} - mc}{\sqrt{m}} \Rightarrow G \quad \text{in } \ell^1, \quad (4.8)$$

and [equation \(1.7\)](#) holds for this general profile. If [equation \(4.7\)](#) fails, the vectors centered at mc are not tight in ℓ^1 either.

Proof. Assume first [equation \(4.7\)](#). By Minkowski's inequality and [equation \(4.4\)](#),

$$\left\| \sum_{j > J} \frac{|K_j^{(m)} - \mu_j(t_m)|}{\sqrt{m}} \right\|_{L^2} \leq C \sum_{j > J} \sqrt{c_j}. \quad (4.9)$$

Thus finite-dimensional convergence upgrades to ℓ^1 convergence. The limiting G belongs to ℓ^1 because $\mathbb{E} \sum_j \sqrt{c_j} |Z_j| < \infty$ and $\sum_j c_j |Z_*| = |Z_*|$.

To replace the centering, use

$$\frac{f(tc_j)}{\sqrt{t}} \leq \min(\sqrt{t} c_j, t^{-1/2}) \leq \sqrt{c_j}.$$

Each term tends to zero. Dominated convergence gives $D(t) = o(\sqrt{t})$, and [equation \(3.6\)](#) then gives $\|\mu(t_m) - mc\|_1 = o(\sqrt{m})$. This proves [equation \(4.8\)](#). Taking $J = -1$ in [equation \(4.9\)](#) supplies uniform second moments of the norms, including the negligible deterministic shift. Thus the norms are uniformly integrable and

$$\lim_m \mathbb{E} \left\| \frac{K^{(m)} - mc}{\sqrt{m}} \right\|_1 = \mathbb{E} \|G\|_1 = \frac{1}{\sqrt{\pi}} \sum_j \sqrt{c_j(1 - c_j)}.$$

For necessity, suppose $\sum \sqrt{c_j} = \infty$. The partial sums of

$$\sum_j \sqrt{c_j} (|Z_j| - \mathbb{E}|Z_0|)$$

form an L^2 -bounded martingale and converge almost surely, since $\sum c_j < \infty$. Therefore $\sum_j \sqrt{c_j} |Z_j| = \infty$ almost surely. Subtracting the summable vector $(c_j Z_*)_j$ cannot change this, so $G \notin \ell^1$ almost surely. If the saddle-centered family were tight in ℓ^1 , a subsequential limit would be an ℓ^1 -valued random vector with the finite-dimensional laws of G , an impossibility.

For centering at mc , tightness of even coordinate zero would force $D(t_m)/\sqrt{m}$ to remain bounded, by [equations \(3.6\)](#) and [\(4.1\)](#). Along a subsequence it would then converge to a finite $d \geq 0$. The finite-dimensional limiting law would be $G + (d/2)c$, since $f(t_m c_j)/\sqrt{m} \rightarrow 0$ for fixed j . Adding this summable deterministic vector still does not give an ℓ^1 vector. The same contradiction proves the last assertion. $\square$

For geometric weights, $\sum_j \sqrt{c_j} = \sqrt{1-q}/(1-\sqrt{q}) < \infty$. The general theorem therefore proves [theorem 1.1](#).

Remark 4.5 (Finite support and zero weights). Zero weights can be discarded. With exactly one positive weight, $K_0^{(m)} = m$ is deterministic. With finitely many positive weights, the proof first observes all but one coordinate and then recovers the last from the total; the full covariance is singular in the expected direction. Every summability condition is automatic. The infinite positive assumption is used to avoid repeating these elementary cases.

5 Geometric saddles and logarithmic phase

Return to $c_j = (1 - q)q^j$. Write

$$b = 1/q > 1, \quad L = \log b, \quad x = t(1 - q), \quad \lambda = \log_b x = n + \theta,$$

where $n = \lfloor \lambda \rfloor$ and $0 \leq \theta < 1$. The sites where tc_j is close to one have j close to n . Their exact rescaled parameters are

$$tc_{n+r} = q^{r-\theta}. \quad (5.1)$$

This identity explains why a fractional logarithm remains in the limit.

Proposition 5.1 (An explicit smooth saddle correction). *Define, initially for $0 \leq \theta \leq 1$,*

$$Q_q(\theta) = 1 - \theta + \sum_{r \leq 0} (f(q^{r-\theta}) - 1) + \sum_{r \geq 1} f(q^{r-\theta}). \quad (5.2)$$

It extends to a smooth one-periodic function. As $t \rightarrow \infty$,

$$D(t) = \log_b(t(1 - q)) + Q_q(\log_b(t(1 - q))) + O_q(te^{-2t(1-q)/q}). \quad (5.3)$$

In particular,

$$t_m = 2m + \log_b(2m(1 - q)) + Q_q(\log_b(2m(1 - q))) + O_q\left(\frac{\log m}{m}\right), \quad (5.4)$$

$$V(t) = \frac{t}{4} + O_q(1). \quad (5.5)$$

Periodic functions are evaluated on real arguments through their periodic extensions.

Proof. For $r \rightarrow +\infty$, the summands in [equation \(5.2\)](#) and every fixed-order θ derivative are $O_q(q^r)$, by [equation \(3.2\)](#). For $r \rightarrow -\infty$, the differences from one and their derivatives decay faster than an exponential in $|r|$, by [equation \(3.3\)](#). The series and all derivatives therefore converge uniformly. Shifting r by one proves $Q_q(\theta + 1) = Q_q(\theta)$, including matching derivatives at the endpoints.

For $n \geq 0$, reindex the exact deficit as

$$D(t) = \sum_{r=-n}^{\infty} f(q^{r-\theta}).$$

Subtract $n + 1$ from the terms with $r \leq 0$ and compare with [equation \(5.2\)](#). The omitted negative-index tail is bounded by a constant times $(x/q)e^{-2x/q}$, since $1 - f(a) = 2a/(e^{2a} - 1)$. This gives [equation \(5.3\)](#) and $D(t) = O_q(\log t)$.

The same estimates after differentiation give $D'(t) = O_q(t^{-1})$. From $t_m - 2m = D(t_m) = O_q(\log m)$, the mean-value theorem gives $D(t_m) = D(2m) + O_q((\log m)/m)$, proving [equation \(5.4\)](#). Finally $M'(t) = 2V(t)/t$ and $M(t) = (t - D(t))/2$ imply $V(t) = t(1 - D'(t))/4$, proving [equation \(5.5\)](#). $\square$

Geometric harmonic sums often produce logarithmic periodicity; Mellin methods provide a broad framework [5]. Here the explicit two-sided series proves everything needed without a contour shift, a zeta-function zero claim, or an imported endpoint expansion.

For the rest of the geometric analysis use the exact definitions

$$n_m = \lfloor \log_b(t_m(1 - q)) \rfloor, \quad \theta_m = \log_b(t_m(1 - q)) - n_m. \quad (5.6)$$

Replacing this floor by an asymptotically close real expression without tracking possible boundary crossings would be incorrect. Our moving-phase statements keep [equation \(5.6\)](#) exactly.

6 The discrete frontier and occupied-count laws

6.1 The complete right-hand tail

Let d_{TV} denote total-variation distance, normalized as one half of the L^1 distance between probability mass functions. For an integer s and a real phase θ , define independent variables

$$Z_r^{(\theta)} \sim B_{a_r(\theta)}, \quad a_r(\theta) = q^{r-\theta}, \quad r \geq s. \quad (6.1)$$

The sum of these variables is finite almost surely. Indeed, their means are $O_q(q^{2r})$ as $r \rightarrow +\infty$.

Theorem 6.1 (Total-variation approximation of the moving tail). *For every fixed integer s ,*

$$d_{\text{TV}} \left(\mathcal{L}((K_{n_m+r}^{(m)})_{r \geq s}), \bigotimes_{r \geq s} \mathcal{L}(Z_r^{(\theta_m)}) \right) \rightarrow 0. \quad (6.2)$$

For sufficiently large m , all indices on the left are nonnegative. Along any subsequence with $\theta_m \rightarrow \theta$, the right-hand product may be replaced by its fixed-phase version at θ .

Proof. Before conditioning, the tail beginning at $n_m + s$ has exactly the product law on the right, by [equation \(5.1\)](#). Denote its total by T_m . For fixed s , $\mathbb{E}T_m$ and $\text{Var} T_m$ are bounded uniformly in m , since $0 \leq \theta_m < 1$.

The head $S_h = S - T_m$ has centered normal limit with variance coefficient $1/4$ on the $\sqrt{t}$ scale. To see this, its difference from the full centered sum is $(T_m - \mathbb{E}T_m)/\sqrt{t} \rightarrow 0$ in L^2 . For large m , the head contains coordinate zero, which supplies the Fourier bound in [lemma 3.2](#). The proof of [theorem 3.3](#) therefore applies to this moving head as well. Its mean is $m - \mathbb{E}T_m$, and its local probabilities at $m - k$, for bounded k , are asymptotic to $\sqrt{2/(\pi t_m)}$ uniformly on such k -sets.

The conditional tail law has density

$$R_m(T_m) = \frac{\mathbb{P}(S_h = m - T_m)}{\mathbb{P}(S = m)}$$

with respect to its independent product law. The local theorem gives $R_m(k) \rightarrow 1$ uniformly for bounded k , and the head's retained coordinate gives a uniform upper bound on R_m . The totals T_m are tight, so $\mathbb{E}|R_m(T_m) - 1| \rightarrow 0$. This proves [equation \(6.2\)](#).

To vary the phase, truncate at a large right endpoint R . On a finite coordinate set the product probabilities depend continuously, in total variation, on θ . The probability of any nonzero coordinate beyond R is $O_q(q^{2R})$ uniformly for $\theta \in [0, 1]$, so the complete products are also continuous in total variation. $\square$

6.2 A two-sided occupancy interface

For $r \in \mathbb{Z}$ define

$$b_r(\theta) = \mathbb{P}(Z_r^{(\theta)} = 0) = \frac{a_r(\theta)}{\sinh a_r(\theta)}. \quad (6.3)$$

We may construct all $Z_r^{(\theta)}$, $r \in \mathbb{Z}$, independently. Their actual count sequence is not summable to the left. Their *deviations from an occupied-left, vacant-right pattern*, however, are finite:

$$\mathcal{D}_\theta(r) = \mathbf{1}_{\{Z_r^{(\theta)} > 0\}} - \mathbf{1}_{\{r \leq 0\}}. \quad (6.4)$$

Indeed,

$$\sum_{r \leq 0} b_r(\theta) + \sum_{r \geq 1} (1 - b_r(\theta)) < \infty, \quad (6.5)$$

uniformly for $\theta \in [0, 1]$. The left tail decreases faster than exponentially in $|r|$, and the right tail is $O_q(q^{2r})$. The first Borel–Cantelli lemma implies finite support almost surely.

For the conditioned model, take m sufficiently large that $n_m \geq 0$ and set

$$\mathcal{D}_m(r) = \begin{cases} \mathbf{1}_{\{K_{n_m+r}^{(m)} > 0\}} - \mathbf{1}_{\{r \leq 0\}}, & r \geq -n_m, \\ 0, & r < -n_m. \end{cases} \quad (6.6)$$

The extension by zero records no artificial vacancies before the original index zero.

Lemma 6.2 (Uniform bounds for any number of omitted sites). *There exists $C_q < \infty$ such that, for all sufficiently large m , every finite set $A \subset \mathbb{N}_0$ with $|A| = p$, and coordinate events $E_j \subseteq \mathbb{N}_0$,*

$$\mathbb{P}(K_j^{(m)} \in E_j \text{ for all } j \in A) \leq C_q q^{-p/2} \prod_{j \in A} \mathbb{P}(B_j(t_m) \in E_j). \quad (6.7)$$

Proof. The complement of A contains some $\ell \in \{0, \dots, p\}$. Its sum has maximal atom at most $C_1(t_m(1-q)q^p)^{-1/2}$ by [equation \(3.9\)](#). Condition on the coordinates in A , use their independence, and divide by [equation \(3.11\)](#). This gives the stated constant independently of A, p , including when the displayed upper bound exceeds one. $\square$

Theorem 6.3 (The full defect field and its functionals). *On the countable space of finitely supported $\{-1, 0, 1\}$ -valued sequences on $\mathbb{Z}$,*

$$d_{\text{TV}}(\mathcal{L}(\mathcal{D}_m), \mathcal{L}(\mathcal{D}_{\theta_m})) \longrightarrow 0. \quad (6.8)$$

Let

$$M_m = \max\{j : K_j^{(m)} > 0\}, \quad H_m = \#\{j : K_j^{(m)} > 0\},$$

and define

$$W_\theta = \sup\{r : Z_r^{(\theta)} > 0\}, \quad L_\theta = -\sum_{r \leq 0} \mathbf{1}_{\{Z_r^{(\theta)} = 0\}} + \sum_{r \geq 1} \mathbf{1}_{\{Z_r^{(\theta)} > 0\}}. \quad (6.9)$$

Then the law of $(M_m - n_m, H_m - (n_m + 1))$ is asymptotic in total variation to the law of $(W_{\theta_m}, L_{\theta_m})$. For $s \in \mathbb{Z}$,

$$\mathbb{P}(W_\theta \leq s) = \prod_{r > s} b_r(\theta), \quad (6.10)$$

and, for real u ,

$$\mathbb{E}e^{uL_\theta} = \prod_{r \leq 0} (1 - b_r + b_r e^{-u}) \prod_{r \geq 1} (b_r + (1 - b_r)e^u). \quad (6.11)$$

All moments and all real exponential moments of $H_m - (n_m + 1)$ converge to the corresponding moving-phase expressions in [equation \(6.11\)](#).

Proof. The case $p = 1$ of [lemma 6.2](#) and a union bound show that the probability of any defect outside $[-R, R]$ is at most a constant times

$$\sum_{r < -R} b_r(\theta_m) + \sum_{r > R} (1 - b_r(\theta_m)),$$

which tends uniformly to zero as $R \rightarrow \infty$. The same bound holds for the independent limiting field by [equation \(6.5\)](#). On $[-R, R]$, [theorem 6.1](#) gives total-variation convergence. Extend these truncated fields by zero and then let $R \rightarrow \infty$; the triangle inequality proves [equation \(6.8\)](#).

For these sufficiently large m , the sum of the field is $H_m - (n_m + 1)$. Its occupied pattern, interpreted as the baseline $\mathbf{1}_{\{r \leq 0\}}$ plus the field, has maximum $M_m - n_m$ for $m \geq 1$; artificial

occupied sites to the far left do not alter that maximum. Thus the joint functional assertion follows by contraction of total variation. Independence of the Z_r proves [equations \(6.10\) and \(6.11\)](#); their products converge by [equation \(6.5\)](#). The maximum is finite almost surely, and the pattern has occupied sites arbitrarily far to the left.

It remains to justify the unbounded moment statements. Let $N_m = \sum_r |\mathcal{D}_m(r)|$, and let $d_{j,m}$ be the corresponding unconditioned defect probability at original site j . [Equation \(6.5\)](#) gives $\sum_j d_{j,m} \leq A_q$ for a finite constant A_q independent of m . Summing [equation \(6.7\)](#) over sets of p defect sites yields

$$\mathbb{E} \binom{N_m}{p} \leq C_q q^{-p/2} \frac{A_q^p}{p!}.$$

For $v \geq 0$, expand $(1 + (e^v - 1))^{N_m}$ to obtain

$$\mathbb{E} e^{v N_m} \leq C_q \exp(q^{-1/2} A_q (e^v - 1)). \quad (6.12)$$

The independent fields obey an analogous uniform bound. Since $|H_m - (n_m + 1)| \leq N_m$, these estimates imply uniform integrability of every power and every fixed real exponential, using a larger exponent in [equation \(6.12\)](#). Total-variation convergence followed by truncation proves the moment assertions. $\square$

Corollary 6.4 (A sharp far-right tail). *For fixed phase, as the integer $s \rightarrow +\infty$,*

$$\mathbb{P}(W_\theta > s) \sim \frac{q^{2(s+1-\theta)}}{6(1-q^2)}. \quad (6.13)$$

The maximum's limit law depends nontrivially on phase.

Proof. As $a \downarrow 0$, $a/\sinh a = 1 - a^2/6 + O(a^4)$. Taking logarithms in [equation \(6.10\)](#) gives the geometric sum in [equation \(6.13\)](#), with error $O_q(q^{4s})$. Phase dependence is strict: increasing θ increases every a_r , and $a/\sinh a$ strictly decreases for $a > 0$, so the CDF in [equation \(6.10\)](#) strictly decreases at every fixed integer s . $\square$

6.3 Mean, variance, and exact phase averages

Define

$$h_q(\theta) = - \sum_{r \leq 0} b_r(\theta) + \sum_{r \geq 1} (1 - b_r(\theta)), \quad v_q(\theta) = \sum_{r \in \mathbb{Z}} b_r(\theta)(1 - b_r(\theta)). \quad (6.14)$$

These are smooth; $h_q(\theta + 1) = h_q(\theta) + 1$ and $v_q(\theta + 1) = v_q(\theta)$. The moment convergence just proved gives

$$\mathbb{E} H_m = n_m + 1 + h_q(\theta_m) + o(1), \quad \text{Var } H_m = v_q(\theta_m) + o(1). \quad (6.15)$$

Proposition 6.5 (Exact averages and mean expansion). *One has*

$$\int_0^1 h_q(\theta) d\theta = -\frac{\log 2}{L}, \quad \int_0^1 v_q(\theta) d\theta = \frac{2 \log 2 - 1}{L}. \quad (6.16)$$

The function

$$P_q(\theta) = h_q(\theta) - \theta + \frac{1}{2} + \frac{\log 2}{L} \quad (6.17)$$

is smooth, one-periodic, and mean zero. Moreover,

$$\mathbb{E} H_m = \log_b((1-q)m) + \frac{1}{2} + P_q(\log_b(2(1-q)m)) + o(1). \quad (6.18)$$

For $q = 1/2$ this specializes to

$$\mathbb{E} H_m = \log_2 m - \frac{1}{2} + P_{1/2}(\log_2 m) + o(1), \quad \int_0^1 v_{1/2}(\theta) d\theta = 2 - \frac{1}{\log 2}. \quad (6.19)$$

Proof. Uniformly convergent tail bounds justify integration term by term. The change of variable $a = q^{r-\theta}$ tiles $(0, \infty)$ as r varies and θ runs from zero to one. Thus

$$L \int_0^1 h_q(\theta) d\theta = \int_0^1 \left(\frac{1}{a} - \frac{1}{\sinh a} \right) da - \int_1^\infty \frac{da}{\sinh a} = -\log 2,$$

using $\int \operatorname{csch} a da = \log \tanh(a/2)$ and the limit $\tanh(a/2) \sim a/2$ at zero. Similarly,

$$L \int_0^1 v_q(\theta) d\theta = \int_0^\infty \left(\operatorname{csch} a - a \operatorname{csch}^2 a \right) da = 2 \log 2 - 1.$$

An antiderivative for the latter integrand is $\log \tanh(a/2) + a \coth a - \log \sinh a$; its limits at infinity and zero are $\log 2$ and $1 - \log 2$.

The shift identity for h_q follows by reindexing the series, and [equation \(6.16\)](#) then proves the assertions about P_q . Substituting [equation \(6.17\)](#) in [equation \(6.15\)](#) gives

$$\mathbb{E}H_m = \log_b(t_m(1-q)) + \frac{1}{2} - \log_b 2 + P_q(\log_b(t_m(1-q))) + o(1).$$

Since $t_m/(2m) = 1 + O_q((\log m)/m)$ and P_q has bounded derivative, this is [equation \(6.18\)](#). $\square$

The variance function is bounded above and bounded away from zero: its series is continuous on the phase circle, and each value contains positive terms. Thus the fluctuations of $H_m - \log_b m$ have order one. They do not have a nondegenerate Gaussian limit after division by $\sqrt{\log m}$. This distinction separates the frontier from the macroscopic Gaussian fluctuations of $K_j^{(m)}$.

6.4 Independence of the two limiting scales

Theorem 6.6 (The bridge and the frontier become independent). *Along every subsequence with $\theta_m \rightarrow \theta$,*

$$\left(\frac{K^{(m)} - mc}{\sqrt{m}}, \mathcal{D}_m \right) \Rightarrow (G, \mathcal{D}_\theta) \quad (6.20)$$

in the product of ℓ^1 and the discrete space of finite defect fields. The two random elements on the right are independent.

Proof. Fix a macroscopic block $I = \{0, \dots, J\}$ and a frontier tail beginning at $n_m + s$. For large m they are disjoint. Before conditioning, the block, the tail, and the intervening remaining coordinates are independent. The tail's total is uniformly tight. The remaining sum has a local limit theorem with variance coefficient $(1 - \sum_{i \in I} c_i)/4$: deleting the tail changes the fixed complement by a variable of bounded variance, and an unused fixed coordinate such as $J + 1$ supplies the Gaussian Fourier bound.

If $z_i = (k_i - \mu_i(t_m))/\sqrt{t_m}$ and the tail has bounded total, the conditional likelihood ratio therefore tends, uniformly for z in compact sets, to

$$\left(1 - \sum_{i \in I} c_i \right)^{-1/2} \exp \left[-\frac{2(\sum_{i \in I} z_i)^2}{1 - \sum_{i \in I} c_i} \right].$$

This limit does not depend on the tail. The ratios are uniformly bounded by the same retained-coordinate estimate. Truncating the tight tail total and then the tight block coordinates proves the product of the finite Gaussian bridge law and the independent tail law.

The defect bounds outside $[-R, R]$ used in [theorem 6.3](#) extend this product convergence to the whole defect field. Finally [equation \(4.9\)](#) extends the finite Gaussian block to ℓ^1 . The deterministic centering difference is $O_q(\log m)/\sqrt{m}$ in ℓ^1 , so the centering by mc in [equation \(6.20\)](#) is valid. $\square$

7 The sharp centering transition for power weights

The geometric theorem might suggest that centering by mc_j works for every summable profile. The exact saddle identity shows that this is false. It also identifies the relevant threshold without changing the random fluctuation mechanism.

Proposition 7.1 (General centering criterion). *For any fixed nonempty finite set of positive coordinates, the centered Gaussian limit in [equation \(4.1\)](#) remains valid when $\mu_j(t_m)$ is replaced by mc_j if and only if*

$$D(t_m) = o(\sqrt{m}). \quad (7.1)$$

If $D(t_m)/\sqrt{m} \rightarrow d < \infty$, the limiting mean is $(d/2)c_j$. If $D(t_m)/\sqrt{m} \rightarrow \infty$, each fixed coordinate $(K_j^{(m)} - mc_j)/\sqrt{m}$ tends to $+\infty$ in probability.

Proof. For fixed j , $f(t_m c_j) \rightarrow 1$. The exact difference [equation \(3.6\)](#) is therefore a deterministic shift of size $c_j D(t_m)/(2\sqrt{m}) + o(1)$ on the fluctuation scale. The saddle-centered sequence converges to a nondegenerate centered normal in that coordinate. The finite-shift and divergent-shift statements follow immediately. Conversely, if [equation \(7.1\)](#) fails, there is a subsequence along which this nonnegative shift is bounded away from zero and has a finite positive or infinite subsequential limit. It cannot have the same centered normal limit. $\square$

Theorem 7.2 (Power-law threshold at exponent two). *Let*

$$c_j = C(j+1)^{-\alpha}, \quad \alpha > 1, \quad C = \zeta(\alpha)^{-1}.$$

Define the positive finite constant

$$A_\alpha = C^{1/\alpha} \int_0^\infty f(x^{-\alpha}) dx. \quad (7.2)$$

Then

$$D(t) = A_\alpha t^{1/\alpha} + O(1). \quad (7.3)$$

For each fixed finite block, the variables centered at mc have the following behavior:

Exponent	Limit on the $\sqrt{m}$ scale
$\alpha > 2$	Centered Gaussian bridge
$\alpha = 2$	Gaussian bridge with mean $c_j A_2/\sqrt{2}$
$1 < \alpha < 2$	Every fixed positive coordinate tends to $+\infty$

The ℓ^1 bridge occurs exactly for $\alpha > 2$. The universal ℓ^2 theorem with saddle centering holds for all $\alpha > 1$.

Proof. The function $g(x) = f(x^{-\alpha})$ is decreasing from one to zero, positive, and integrable: it is bounded near zero, and $g(x) = O(x^{-\alpha})$ at infinity. Monotonicity follows either by direct differentiation or from

$$f'(a) = \frac{2((2a-1)e^{2a}+1)}{(e^{2a}-1)^2} > 0.$$

Set $s = (Ct)^{1/\alpha}$. Then $D(t) = \sum_{n \geq 1} g(n/s)$. Comparison of a right Riemann sum with its integral gives

$$0 \leq s \int_0^\infty g(x) dx - \sum_{n \geq 1} g(n/s) \leq 1.$$

This proves [equation \(7.3\)](#), including its bounded error. Since $t_m \sim 2m$, the three regimes follow from [proposition 7.1](#). Finally $\sum_j \sqrt{c_j} < \infty$ holds exactly when $\alpha > 2$; apply [theorem 4.4](#). $\square$

Proposition 7.3 (Optional evaluation of the power-law constant). *With $\beta = 1/\alpha \in (0, 1)$,*

$$A_\alpha = -\beta(2C)^\beta \Gamma(1 - \beta) \zeta(1 - \beta) > 0. \quad (7.4)$$

In particular,

$$A_2 = -\sqrt{\frac{\pi C}{2}} \zeta(1/2). \quad (7.5)$$

Proof. Put $F(y) = (e^y - 1)^{-1} - y^{-1}$. For $0 < s < 1$, the integral $I(s) = \int_0^\infty F(y) y^{s-1} dy$ is absolutely convergent. The identity $F(y) - 2F(2y) = (e^y + 1)^{-1}$ gives

$$(1 - 2^{1-s})I(s) = \int_0^\infty \frac{y^{s-1}}{e^y + 1} dy = \Gamma(s)\eta(s),$$

where $\eta(s) = \sum_{n \geq 1} (-1)^{n-1} n^{-s}$. For completeness, expansion of $(e^y + 1)^{-1}$ into alternating exponentials is justified in the integral by its alternating remainder: the integral of the remainder is at most $\Gamma(s)(N+1)^{-s}$. The standard identity $\eta(s) = (1 - 2^{1-s})\zeta(s)$, or equivalently this definition of the continued zeta function on $(0, 1)$, gives $I(s) = \Gamma(s)\zeta(s)$. Substitute $u = x^{-\alpha}$ in [equation \(7.2\)](#), then put $y = 2u$ and use $f(u) = -2uF(2u)$. This gives $A_\alpha = -\beta(2C)^\beta I(1 - \beta)$, proving the formula. Positivity already follows from [equation \(7.2\)](#). $\square$

The positive integral in [equation \(7.2\)](#) is sufficient for the threshold theorem. The special-function evaluation adds an exact constant, not an additional assumption.

8 A full large deviation principle in ℓ^1

The local and Gaussian theorems describe typical fluctuations. A different exact identity determines the exponential cost of an atypical profile. This section applies to every profile in [equation \(2.1\)](#). Let

$$\mathcal{P} = \{p \in \ell^1 : p_j \geq 0, \sum_j p_j = 1\}, \quad P_m = K^{(m)}/m.$$

All logarithms in relative entropy are natural logarithms. Define

$$D(p||c) = \sum_j p_j \log \frac{p_j}{c_j} = \sum_j c_j \phi(p_j/c_j), \quad \phi(x) = x \log x - x + 1, \quad (8.1)$$

with $0 \log 0 = 0$ and value $+\infty$ allowed. The nonnegative second series gives an unambiguous definition even when the entropy is infinite.

Theorem 8.1 (Twice-relative-entropy large deviations). *The laws of P_m satisfy the full large deviation principle on $\mathcal{P}$ with its ℓ^1 topology, at speed m , with good rate*

$$I(p) = 2D(p||c). \quad (8.2)$$

Explicitly, for every relatively open $G \subseteq \mathcal{P}$ and relatively closed $F \subseteq \mathcal{P}$,

$$\liminf_{m \rightarrow \infty} \frac{1}{m} \log \mathbb{P}(P_m \in G) \geq - \inf_{p \in G} I(p), \quad (8.3)$$

$$\limsup_{m \rightarrow \infty} \frac{1}{m} \log \mathbb{P}(P_m \in F) \leq - \inf_{p \in F} I(p). \quad (8.4)$$

The sublevel sets of I are compact in ℓ^1 .

We give the proof in several steps, including exponential tightness, so that no change between a coordinate topology and ℓ^1 is implicit.

8.1 An exact exponential-tilt identity

For any nonnegative summable profile a , not necessarily of total mass one, put $\mu_{2m}(a) = \mathbb{E}(\sum_j a_j U_j)^{2m}$. By independence and symmetry of finite sums and then monotone convergence of their nonnegative coefficient expansions,

$$\mu_{2m}(a) = (2m)! \sum_{\sum_j k_j = m} \prod_j \frac{a_j^{2k_j}}{(2k_j + 1)!}. \quad (8.5)$$

Convergence of the random sums is uniform almost surely, so their moments converge to the same expression.

If $\vartheta = (\vartheta_j)$ is any real sequence with $A(\vartheta) = \sum_j c_j e^{\vartheta_j/2} < \infty$, then Tonelli's theorem applied to [equation \(8.5\)](#) proves

$$\mathbb{E} e^{\sum_j \vartheta_j K_j^{(m)}} = \frac{\mu_{2m}((c_j e^{\vartheta_j/2})_j)}{\mu_{2m}(c)}. \quad (8.6)$$

This holds for unbounded tilts as well as bounded ones; the summability of the tilted profile is the needed hypothesis.

For a positive total mass $A = \sum_j a_j$, compact support gives $|\sum a_j U_j| \leq A$. Conversely, for every $\varepsilon > 0$ there is positive probability that this sum exceeds $A - \varepsilon$: choose a finite head with tail mass less than $\varepsilon/4$ and require each of its finitely many uniform digits to be sufficiently close to one. It follows that

$$\mu_{2m}(a)^{1/(2m)} \longrightarrow A. \quad (8.7)$$

Thus [equation \(8.6\)](#) gives

$$\Lambda(\vartheta) := \lim_{m \rightarrow \infty} \frac{1}{m} \log \mathbb{E} e^{m \langle \vartheta, P_m \rangle} = 2 \log \sum_j c_j e^{\vartheta_j/2}. \quad (8.8)$$

No estimate on how quickly the density vanishes at its endpoint enters this argument.

8.2 Finite partitions and entropy duality

For $J \geq 0$, collapse all coordinates above J into a single tail cell:

$$p^{[J]} = (p_0, \dots, p_J, p_{>J}), \quad c^{[J]} = (c_0, \dots, c_J, c_{>J}).$$

The log-moment formula for this finite simplex is

$$\Lambda_J(\vartheta) = 2 \log \left(\sum_{j=0}^J c_j e^{\vartheta_j/2} + c_{>J} e^{\vartheta_{>J}/2} \right).$$

Its convex dual is $I_J(p^{[J]}) = 2 D(p^{[J]} \| c^{[J]})$. For positive coordinates the optimizing tilt is $\vartheta_i = 2 \log(p_i / c_i^{[J]})$, up to adding a common constant. Zero coordinates follow by limits.

The finite-simplex large deviation bounds can be proved directly. For an interior target p , tilt by the preceding ϑ . The new limiting logarithmic moment generating function is $\Lambda_J(\vartheta + u) - \Lambda_J(\vartheta)$, with gradient p at zero. Chernoff bounds for each coordinate, using differentiability at zero, show that the tilted law concentrates exponentially in every neighborhood of p . Changing back to the original law gives lower exponent $-\langle \vartheta, p \rangle + \Lambda_J(\vartheta) = -I_J(p)$. For boundary targets, approach by $(1 - \varepsilon)p + \varepsilon c^{[J]}$; finite-simplex entropy is continuous. Chernoff bounds followed by a finite cover of the compact simplex give the upper bound. This proves both finite-dimensional bounds without presuming nonsingularity in the direction of the total mass.

Jensen's inequality applied to ϕ in [equation \(8.1\)](#) gives

$$I_J(p^{[J]}) \uparrow I(p). \quad (8.9)$$

Indeed each finite partition entropy is at most the full entropy, whereas its nonnegative sum over the first $J + 1$ cells already increases to the full sum. Consequently

$$I(p) = \sup_{\vartheta \in \ell^\infty} \{\langle \vartheta, p \rangle - \Lambda(\vartheta)\}. \quad (8.10)$$

For the upper inequality, apply nonnegativity of entropy to the tilted probability profile $c_j e^{\vartheta_j/2} / A(\vartheta)$; for the reverse inequality optimize over tilts constant on each finite partition and use [equation \(8.9\)](#). Formula [equation \(8.10\)](#) also proves lower semicontinuity in ℓ^1 .

8.3 Explicit exponential tightness

Let $T_j = \sum_{k \geq j} c_k$ and

$$w_j = -\frac{1}{2} \log T_j. \quad (8.11)$$

Then $w_j \uparrow \infty$, $w_0 = 0$, and

$$A := \sum_j c_j e^{w_j} = \sum_j \frac{c_j}{\sqrt{T_j}} \leq 2. \quad (8.12)$$

To verify the bound, use $c_j = T_j - T_{j+1}$ and

$$\frac{c_j}{\sqrt{T_j}} = (\sqrt{T_j} - \sqrt{T_{j+1}}) \left(1 + \sqrt{T_{j+1}/T_j}\right) \leq 2(\sqrt{T_j} - \sqrt{T_{j+1}})$$

and telescope. The unbounded tilt $\vartheta_j = 2w_j$ is therefore admissible in [equation \(8.8\)](#). Markov's inequality yields

$$\limsup_m \frac{1}{m} \log \mathbb{P} \left(\sum_j w_j P_{m,j} > R \right) \leq -2R + 2 \log A \leq -2R + 2 \log 2. \quad (8.13)$$

The sets

$$\mathcal{K}_R = \{p \in \mathcal{P} : \sum_j w_j p_j \leq R\}$$

are compact in ℓ^1 : they are closed, and their tails are uniformly bounded by $R/w_{J+1} \rightarrow 0$, so a coordinatewise diagonal subsequence converges in ℓ^1 . This proves exponential tightness in the required topology.

8.4 Completion of the infinite-dimensional bounds

Fix p in an ℓ^1 -open set G . Choose a ball of radius ε about p inside G , and take J with $2p_{>J} < \varepsilon/2$. If another probability profile q satisfies $\sum_{j \leq J} |q_j - p_j| < \delta$, then

$$\|q - p\|_1 \leq 2\delta + 2p_{>J}.$$

For small δ , this finite-coordinate cylinder lies inside G . The finite-partition lower bound gives exponent at least $-I_J(p^{[J]}) \geq -I(p)$. Optimizing over $p \in G$ proves [equation \(8.3\)](#).

For a compact set $C \subset \mathcal{P}$ and a real $a < \inf_{p \in C} I(p)$, [equation \(8.10\)](#) supplies for each $p \in C$ a bounded tilt with $\langle \vartheta, p \rangle - \Lambda(\vartheta) > a$. This strict inequality holds on a neighborhood of p . A Chernoff bound gives upper exponent at most $-a$ there. A finite subcover gives the compact upper bound. For a closed F , apply it to $F \cap \mathcal{K}_R$ and use [equation \(8.13\)](#) on the complement; letting $R \rightarrow \infty$ proves [equation \(8.4\)](#).

Finally, use [equation \(8.10\)](#) with bounded truncations of $2w_j$ to obtain

$$I(p) \geq 2 \sum_j w_j p_j - 2 \log A.$$

Every rate sublevel set is therefore contained in a compact $\mathcal{K}_R$. Lower semicontinuity proves that it is compact. This completes the proof of [theorem 8.1](#).

Corollary 8.2 (Exponential concentration and a strong law). *For every $\varepsilon > 0$,*

$$\limsup_m \frac{1}{m} \log \mathbb{P}(\|K^{(m)}/m - c\|_1 \geq \varepsilon) \leq -\varepsilon^2. \quad (8.14)$$

Consequently $K^{(m)}/m \rightarrow c$ in ℓ^1 almost surely under any common coupling of the occupancy vectors. Independence between levels is not required.

Proof. The case $p = c$ is immediate. Otherwise, the entropy inequality $2D(p||c) \geq \|p - c\|_1^2$ follows by collapsing to $A = \{j : p_j > c_j\}$ and its complement, where $0 < c(A) < 1$. For a fixed reference probability, binary relative entropy has second derivative $1/(x(1-x)) \geq 4$, so its value is at least twice the square of the difference of the binary probabilities. That difference is $\|p - c\|_1/2$. Apply the large deviation upper bound to the closed set in [equation \(8.14\)](#). For each fixed $\varepsilon > 0$ the probabilities are summable, and Borel–Cantelli, followed by countably many ε , proves the strong law. $\square$

The quadratic expansion of $2D(p||c)$ about c is consistent with the half-multinomial Gaussian covariance. The Gaussian theorem was proved separately: differentiating a limiting logarithmic moment generating function alone would not have justified it.

9 Reproducible exact and numerical checks

The companion program `verify_occupancy.py` uses exact rational arithmetic for small levels and positive recurrences in logarithmic arithmetic for larger levels. It uses no Monte Carlo. All tables and figures in this section are reproducible from the included source and CSV files. The numerical calculations test finite identities and illustrate limits; the preceding proofs establish the asymptotic statements.

9.1 A positive moment recurrence and an exact marginal

For geometric weights, $Y_q \stackrel{d}{=} (1-q)U + qY'_q$, independently. Write $\mu_r = \mathbb{E}Y_q^{2r}$. Expanding the even moment and isolating the self-term gives

$$(1 - q^{2m})\mu_m = \sum_{k=1}^m \binom{2m}{2k} \frac{(1-q)^{2k} q^{2(m-k)}}{2k+1} \mu_{m-k}, \quad \mu_0 = 1. \quad (9.1)$$

Here and only in this computational subsection the subscript m on μ_m denotes the $2m$ th moment. Coefficient extraction gives

$$\mathbb{P}_m(K_0 = k) = \binom{2m}{2k} \frac{(1-q)^{2k} q^{2(m-k)} \mu_{m-k}}{(2k+1)\mu_m}, \quad 0 \leq k \leq m. \quad (9.2)$$

In particular $\mathbb{P}_m(K_0 = 0) = q^{2m}$ exactly. Conditional on $K_0 = k$, the remaining coordinates, reindexed from zero, have the same occupancy law with total $m - k$. This follows because

multiplying every tail weight by the same q contributes a common factor for the fixed residual total.

This self-similarity computes the covariance of K_0, K_1 without truncating the infinite digit set. It also yields an exact residual-total chain for computing the expected ℓ^1 error and a recursion for the occupied count. For the latter, empty first sites can be skipped: conditional on $K_0 > 0$, the occupied count is $1 + H_{m-K_0}$, and the empty-site case has the original H_m law. Solving for the mean and variance gives the recurrences used by the program.

The exact checks use `fractions.Fraction` at $q = 1/2$ through $m = 12$. They verify positivity, normalization, the vacancy identity, direct two-head coefficient extraction, the conditional self-similarity, the covariance formula, and monotonicity of exact maximum-index CDFs.

Table 1: Dyadic allocations computed from the positive moment recurrence. The final row gives the limits from the bridge theorem.

m	$\mathbb{E}K_0/m$	$\text{Var}(K_0)/m$	$\text{Cov}(K_0, K_1)/m$
16	0.5587271132	0.1416368701	-0.0766758432
64	0.5218299035	0.1309863936	-0.0669174041
256	0.5073515386	0.1269729311	-0.0638397003
1024	0.5023213920	0.1256142933	-0.0628952716
Limit	1/2	1/8	-1/16

Remark 9.1 (Why moment comparisons have the stated limits). The finite-coordinate CLT also permits second-moment convergence here. For fixed j , logarithmic differentiation of [equation \(3.7\)](#), using the large- a expansion, gives $\mathbb{E}(B_a - \mu(a))^4 = O(a^2)$ as $a \rightarrow \infty$. The marginal domination bound transfers this to $\mathbb{E}(K_j - \mu_j(t_m))^4 = O(m^2)$. Thus the squared normalized coordinates and their products are uniformly integrable. Their limiting variances and covariances are indeed those in [equation \(1.6\)](#), not merely suggestive quantities extracted from a weak limit.

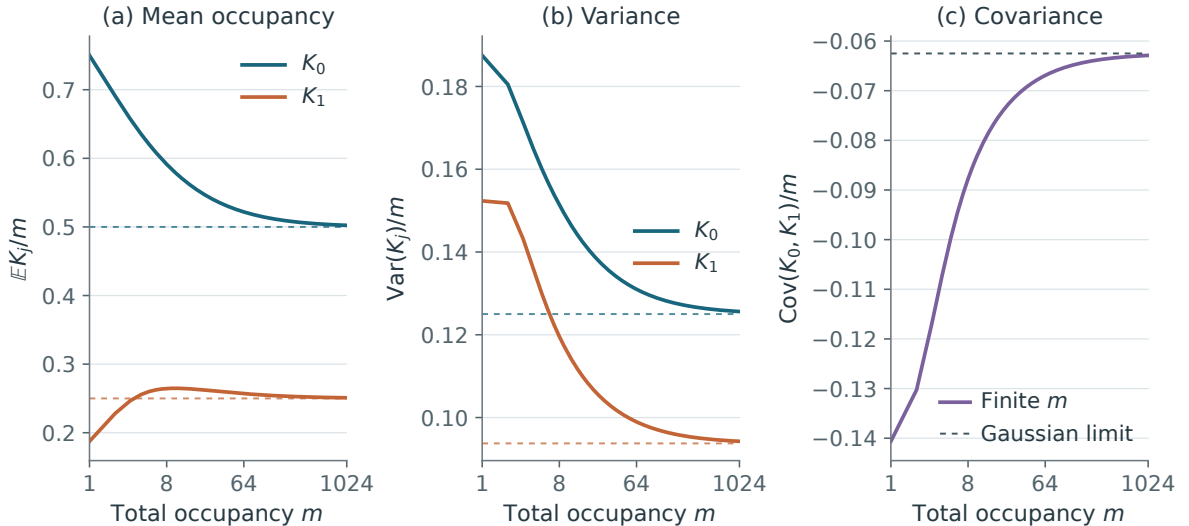


Figure 1: The dyadic mean profile, variances, and covariance through $m = 1024$. Solid curves use the finite moment recurrence; dashed lines show the proved limiting constants.

9.2 The global error constant

For $q = 1/2$, [equation \(1.7\)](#) gives

$$C_{1/2} = \frac{1}{\sqrt{\pi}} \sum_{j \geq 0} \sqrt{2^{-j-1}(1 - 2^{-j-1})} \approx 1.187645633962268. \quad (9.3)$$

The omitted positive tail in this displayed numerical evaluation is at most 8.56×10^{-16} ; this does not include floating-point rounding. The finite values of $\sqrt{m} \mathbb{E} \|K^{(m)}/m - c\|_1$ are 1.2014456933, 1.2084721023, 1.2006282002, and 1.1936305171 for $m = 16, 64, 256, 1024$, respectively. Convergence need not be monotone. Their omitted coordinate tails, after this scaling, are less than 10^{-11} .

To bound those omitted tails, let R_{J+1} be the total remaining after sites $0, \dots, J$ in the exact residual-total chain. Then

$$\sum_{j > J} \mathbb{E} |K_j - mc_j| \leq \mathbb{E} R_{J+1} + mq^{J+1}.$$

This deterministic inequality controls truncation without treating a small spatial digit coefficient as a bound on an occupancy probability.

9.3 Frontier checks and finite occupied counts

The saddle is computed from the exact infinite-sum equation, not replaced by $2m$. At $q = 1/2$ and $m = 1024$ it is

$$t_m \approx 2059.508071235173, \quad n_m = 10, \quad \theta_m \approx 0.008084064817.$$

The finite maximum distribution is computed by extracting the degree- m coefficient from a finite head product and dividing by the full coefficient in [equation \(2.3\)](#). The resulting errors against [equation \(6.10\)](#) are shown in [table 2](#).

Table 2: Maximum difference between the finite maximum-index CDF and its moving-phase approximation on the tested integer range $s = -3, \dots, 5$. These are measured errors on that range, not global error bounds.

m	Maximum absolute CDF difference
16	0.012993410556
64	0.004041664656
256	0.001110769695
1024	0.000285608694

At $m = 1024$, the exact finite recurrences give

$$\mathbb{E} H_m \approx 9.508421724209, \quad \text{Var } H_m \approx 0.556160513709.$$

The moving-phase predictions in [equation \(6.15\)](#) are 9.508071235173 and 0.557174668735, respectively. Numerically integrating the limiting functions over phase gives -1 for the mean correction and 0.557304959111037 for the variance, agreeing with the exact expressions in [equation \(6.16\)](#).

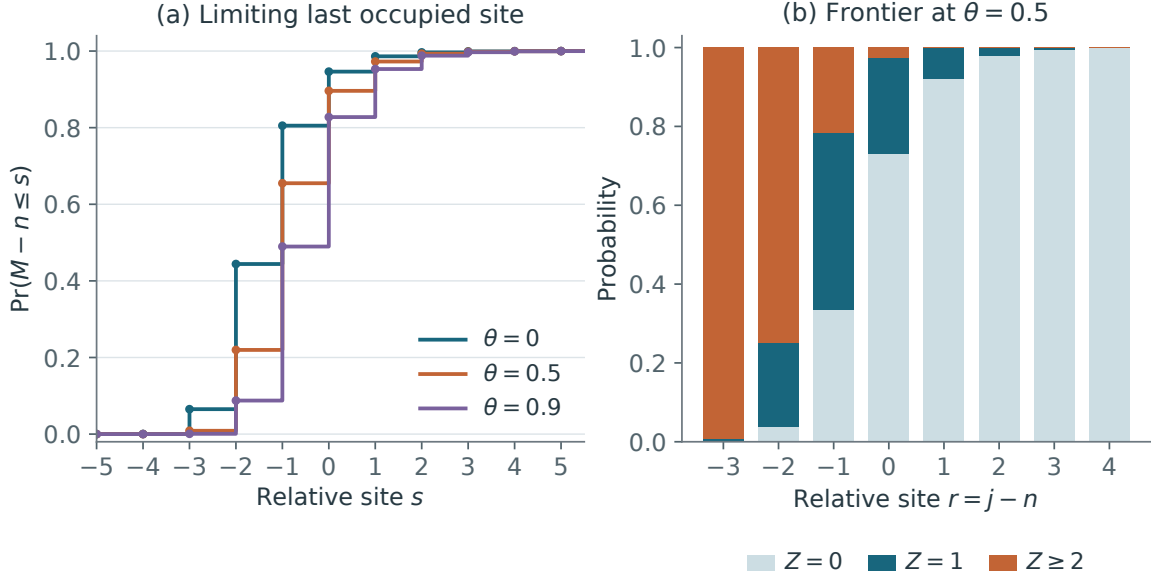


Figure 2: The discrete geometric frontier at $q = 1/2$. Left: the limiting maximum CDF for three phases, showing why the phase must be retained. Right: the limiting occupancy probabilities near the frontier for phase $\theta = 1/2$. These panels display the explicit limiting laws, not sampled finite configurations.

9.4 Numerical reliability and limitations

The program's verification record reports maximum relative disagreement 5.78×10^{-15} between small exact moments and their floating evaluations, maximum raw marginal normalization defect 4.00×10^{-15} , and maximum absolute saddle-equation residual 4.41×10^{-13} for the selected levels. The omitted saddle tail is bounded by 2.00×10^{-15} , and the omitted logarithmic maximum-CDF tail by 1.98×10^{-16} . These analytic tail bounds do not enclose all rounding or quadrature errors; the computation is not interval arithmetic.

The tail bounds follow from elementary inequalities

$$\mu(a) \leq \frac{a^2}{6}, \quad 0 \leq \log \frac{\sinh a}{a} \leq \frac{a^2}{6}. \quad (9.4)$$

For the first, compare positive power-series coefficients in $a \cosh a - \sinh a \leq (a^2/3) \sinh a$; the coefficient inequality at degree $2k + 1$ reduces to $3 \leq 2k + 1$ for $k \geq 1$. Divide by $2 \sinh a$. The second follows by integrating $\frac{d}{da} \log(\sinh a/a) = 2\mu(a)/a \leq a/3$. Summing the resulting geometric tails gives the bounds used in the code.

10 Further research questions and formalization plan

The selected occupancy conjecture is proved in [theorem 1.1](#). The following questions concern stronger, different, or more uniform statements. They are not assumptions in any preceding proof.

Question 10.1 (Effective local and global error bounds). Can the uniform local limit theorem be made effective with explicit constants in q and m , and can the frontier total-variation error be bounded by C_q/m or $C_q(\log m)/m$? The numerical CDF errors in [table 2](#) suggest an inverse- m scale on the tested range, but do not establish a global rate. A useful first target is a third- and fourth-cumulant Fourier estimate that remains uniform after deleting a bounded-mean tail.

Question 10.2 (An Edgeworth expansion for the bridge). Determine the first nonzero correction to the finite Gaussian bridge law, with the logarithmic saddle displacement and its phase retained. Then determine which weighted sequence norms allow a corresponding expansion. An expansion of the total normalizing coefficient alone is insufficient: the marked coordinate transforms and the conditioning likelihood must be expanded together.

Question 10.3 (Quantitative independence of bulk and frontier). Find an explicit distance between the joint law in [equation \(6.20\)](#) and the product of its two limits. A particularly concrete observable is the covariance between a fixed centered occupancy and $H_m - \log_b m$. The theorem proves asymptotic independence in distribution but does not give its first finite- m interaction term.

Question 10.4 (The dense-scale limit $q \uparrow 1$). The fixed- q frontier variance has phase average $(2 \log 2 - 1)/\log(1/q)$, which diverges as $q \uparrow 1$. What are the joint limits when $q = q_m \uparrow 1$ with m ? One expects a transition from a bounded discrete frontier to an approximately continuous occupied-count fluctuation. The regime of validity of the retained-coordinate Fourier bound must be checked as $c_0 = 1 - q_m$ tends to zero.

Question 10.5 (Nongeometric moving fronts). For regularly varying weights, the number of sites with $t_m c_j$ of order one grows with m , and the geometric lattice-phase description changes. Determine the correct point-process or empirical-field limit, its coupling to the ℓ^2 bridge, and the fluctuations of the number of occupied sites. The centering transition in [theorem 7.2](#) gives one constraint that any such result must respect.

Question 10.6 (Other symmetric digit laws). Replace the uniform digits by symmetric compactly supported variables with known endpoint behavior. The moment allocation formula in [equation \(2.4\)](#) survives, but the elementary hyperbolic local factors are replaced by their even moment transforms. Which endpoint hypotheses preserve amplitude limit shapes, Gaussian bridge covariances, and entropy rates? Identify conditions that exclude extra saddles or failure of a span-one local limit theorem.

Question 10.7 (Other factorial residue classes). For allocation weights involving $(dk + r)!$ in place of $(2k + 1)!$, with $d \geq 2$, derive an analogue using Poisson variables conditioned to a residue class. The present proof suggests covariance scaled by $1/d$ and entropy rate $d D(p||c)$ in suitable settings. Proving a uniform Fourier estimate at all lattice peaks, especially with mixtures of residue classes, is the necessary next step; no such general theorem is claimed here.

Question 10.8 (Exact sampling with a certified infinite tail). The conditioning probability is asymptotic to $1/\sqrt{\pi m}$. Can a sampler exploit the independent representation, the deterministic geometric tail, and the explicit frontier to obtain exact samples with near-linear or better expected work in the effective number of sites? An approximate sampler should report a total-variation error for the occupancy vector itself. A small omitted coefficient in the underlying real random series is not such a certificate.

Question 10.9 (Moderate deviations and sharp entropy asymptotics). Between the $\sqrt{m}$ Gaussian scale and the speed- m large deviation principle, establish a moderate deviation principle in sequence spaces. For fixed atypical profiles, derive the polynomial prefactor and the boundary corrections to $\exp(-2m D(p||c))$. Infinite-entropy profiles and profiles with a growing number of occupied coordinates will require separate uniformity arguments.

Question 10.10 (The remaining zero-bias conjectures). The source report also proposes spectral non-reappearance, a Bessel–Airy zero transition, all-order phase-resolved Gaussianization of the transformed real variable, and monotonicity of a standardized Stein discrepancy. They concern different observables. Resolving the allocation conjecture does not resolve those four statements. It would be useful to determine which can use the joint bridge/frontier representation and which require independent analytic input.

10.1 A concrete formalization sequence

The present proof separates into reusable layers. A Lean implementation could proceed in the following order.

1. **Finite algebra.** Formalize the local hyperbolic series, the factorial identity, the finite coefficient normalization, and the finite conditional probability formula. These are suitable for exact checks before any asymptotic library is needed.
2. **Countable probability.** Construct independent B_a laws; prove almost-sure finiteness of $S(t)$, positivity of the conditioning event, and passage from finite coefficients to [equation \(2.6\)](#).
3. **Scalar analysis.** Establish the moment formulas, variance bounds, the unique saddle, and the global characteristic-function bound.
4. **Fourier inversion.** Prove the lattice local limit theorem using an L^1 dominated-convergence lemma for rescaled characteristic functions. This is the key reusable asymptotic component.
5. **Sequence-space convergence.** Combine finite block conditioning with the uniform omitted-coordinate bound, first in ℓ^2 , then in ℓ^1 , and formalize the Gaussian necessity argument for square-root summability.
6. **Geometric and entropy layers.** Add the moving-tail total-variation theorem, defect-field moment bounds, and the separate large deviation proof through finite partitions and explicit compact sets.

No code in this package is represented as a Lean proof. The numerical script and its rational assertions verify finite examples, while the article specifies the mathematical obligations for a future formalization.

A Repository provenance and reproduction

The source report is linked in [\[1\]](#). It lies in the `Fabius_Zero_Bias_Frontier_Report` directory under the representations drafts in `Analysis/FabiusFunction/docs`. At the inspected repository commit `a21208b3ff14a07a4c8318dbf916d543acbef043`, its Git blob identifier is `3e406ee42205a5ea223863c2bfb4c0a8cece08a`. The exact labels used to identify the problem are `eq:q-occupancy`, `eq:occupancy-partition`, `conj:occupancy-shape`, `eq:occupancy-LLN`, and `eq:occupancy-CLT`. The accompanying source README explicitly distinguishes the report’s paper-level claims from the formal Lean corpus.

The package contains this self-contained LaTeX source and its compiled PDF, the numerical program, dependency requirements, numerical notes, the CSV/JSON records, and the two figures in PDF and PNG form. No third-party article or repository source is redistributed.

To regenerate the computations, run from the extracted package:

```
python -m pip install -r requirements.txt
python verify_occupancy.py
```

The recorded run used Python 3.12.14, NumPy 2.3.5, SciPy 1.17.0, and Matplotlib 3.10.8. The program records its actual runtime versions in `data/verification_summary.json`. The provided requirements specify compatible minimum versions; reproducing the recorded version set is preferable when comparing the last printed digits. Exact rational identities are independent of floating-point versions.

To rebuild the article, keep the supplied `figures/` directory beside the source and run:

```
pdflatex -interaction=nonstopmode -halt-on-error occupancy_article.tex
pdflatex -interaction=nonstopmode -halt-on-error occupancy_article.tex
pdflatex -interaction=nonstopmode -halt-on-error occupancy_article.tex
```

The bibliography is embedded in the source. It requires no BibTeX run, network access, or repository-specific notation file.

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